Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - I
Course Code : CSE 0533 1101 (CSE) & PHY 0533 1101 (EEE)

Term: Spring – 2024	Date: 15.01.2024	Time: 10.00 AM	Room: 327	Activity: Lecture 5

Topics to be discussed: Electric potential, relation between electric potential and electric field

Target CLO & PLO : CLO2-3 & PLO 2 & 5

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Explain the concept of electric potential
- ❖ Define equipotential surface
- ❖ Describe what will be the potential due to a point charge and the system of charge.
- ❖ Relate electric potential and electric field
- ❖ Drive the relation of electric potential energy
- ❖ Solve some recommended mathematical problems

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

1) 1 Volt =?
   a) 1 Coulomb b) 1 Newton c) 1 Newton/1 Coulomb d) 1 Joule/1 Coulomb e) 1 Newton/1 meter
2) Potential at all points on the surface of a conductor is –
   a) same. b) not same. c) zero. d) infinite. e) double to the core.

Sample Questions (SAQ):

1) What do you mean electric potential and electric potential energy?
2) Distinguish between electric potential and electric potential energy.
3) Define equipotential surface. Draw the diagram of equipotential surfaces for an isolated positive point charge.
4) Show that the electric field strength at a point in an electric field is equal to the negative gradient of the potential at the same point.
5) Calculate the electric potential energy of a system of charges.
6) What must the magnitude of an isolated positive point charge be for the electric potential at 10 cm from the charge to be 100 volts?
7) What is the electric potential at the surface of a gold nucleus? The radius is $6.6 \times 10^{-15} m$ and the atomic number $Z = 79$.
8) Three charges $\mathrm{q}_1 = -5 \times 10^{-8}\mathrm{coul}, \mathrm{q}_2 = 2 \times 10^{-8}\mathrm{coul}, \mathrm{q}_3 = 3 \times 10^{-8}\mathrm{coul}$ are placed respectively at the three corners of an equilateral triangle; the length of any side ‘a’ of it is 10 cm. Calculate the potential energy of the system.

Reference Books:
1. Giasuddin ahmed – Physics for Engineers (Part-2)
2. Huq. Rafiqullah- Electricity and Magnetism

Pre-reading book for next Lecture:
- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 343-358
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 37-86